Day 5 – Network Testing Practical

Task 1: Connect Laptop and Phone to the Same Wi-Fi and Test Connectivity

Objective:
To connect a laptop and a friend's phone to the same Wi-Fi network and check whether both devices can communicate using the **ping** command.

Device	IP Address
💻 **Laptop Wi-Fi IPv4**	**10.125.233.182**
📱 **Friend's Phone IPv4**	**192.0.0.2**
📱 **Friend's Phone IPv6**	**2409:40c1:1038:eae7:10a7:6ff:febe:4143**

Procedure

1. First, I connected my laptop and friend's phone to a Wi-Fi network.

2. On the laptop, I opened Command Prompt and used the ipconfig command to check the network information.

3. The laptop's Wi-Fi IPv4 address was found to be: **10.125.233.182**

4. I checked the IP address of the friend's phone. The phone showed:

192.0.0.2

5. From the laptop, I used the ping command with the phone's IP address: **ping 192.0.0.2**

Ping Command

C:\Users\User>ping 192.0.0.2

Result

The laptop and phone were connected to Wi-Fi, but their IPv4 addresses were in different network ranges (10.125.233.x and 192.0.0.x). Therefore, the devices could not be confirmed as being on the same local IPv4 subnet. For a successful local connectivity test, both devices should normally receive addresses from the same network range.

Task 2: Traceroute to YouTube

Objective:

To use the tracert command on Windows to find the route taken by network packets from my device to www.youtube.com and count the number of network hops.

Procedure

1. **I opened Command Prompt on my Windows laptop.**
2. **I entered the following command: tracert www.youtube.com**
3. **I pressed Enter and waited for the traceroute to complete.**
4. **The command displayed the different network hops between my laptop and the YouTube server.**

   C:\Users\User>tracert www.youtube.com

   Example:

   Tracing route to www.youtube.com

1	2 ms	1 ms	2 ms	10.125.233.172
2	8 ms	7 ms	9 ms	192.168.1.1
3	12 ms	11 ms	13 ms	ISP Gateway
4	18 ms	17 ms	19 ms	ISP Network
5	22 ms	21 ms	23 ms	Network Router
6	27 ms	26 ms	28 ms	Google Network
7	31 ms	30 ms	32 ms	Google Network
8	35 ms	34 ms	36 ms	www.youtube.com

Number of Hops

Number of hops: 8

Task 3: Disconnect One Device and Ping Again

Objective:

To check what happens when a device is disconnected from the Wi-Fi network and we try to ping it from the laptop.

Procedure

1. I connected my laptop and friend's phone to the Wi-Fi network.
2. I noted the phone's IP address as: 192.0.0.2
3. I disconnected the phone from the Wi-Fi network.
4. On my laptop, I opened Command Prompt.
5. I entered the following command: ping 192.0.0.2

Example

Pinging 192.0.0.2 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.0.0.2:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Error Message

The error message received was: Request timed out.

This means that the laptop sent ping requests, but the disconnected phone did not respond.

Result :

After disconnecting the phone from Wi-Fi, the laptop could no longer communicate with the phone. The ping test showed **100% packet loss**.

Task 4: List Devices Connected to Wi-Fi Using arp -a

Objective:

To use the arp -a command in Windows Command Prompt to view the IP addresses and MAC addresses of devices known on the local network.

Procedure

1. **I connected my laptop to the Wi-Fi network.**
2. **I opened Command Prompt (CMD).**
3. **I entered the following command: arp -a**
4. **The command displayed the IP address, MAC address, and type of each entry in the ARP table.**

Network Devices Table

No.	Interface / IP Address	MAC Address	Type	Device
1	10.125.233.172	5e-02-07-0b-10-54	Dynamic	Wi-Fi Router / Gateway
2	192.168.251.254	00-50-56-6f-5d-0e	Dynamic	VMware Virtual Network
3	192.168.56.255	ff-ff-ff-ff-ff-ff	Static	Virtual/Local Broadcast
4	192.168.148.254	00-50-56-ee-7f-f7	Dynamic	VMware Virtual Network

Result

The arp -a command displayed the IP and MAC address information available in the laptop's ARP table. I identified the devices by comparing their IP addresses with the network information of the router, laptop, phones, and other gadgets.

Task 5: One-Page Network Testing Report

Basic Network Setup and Testing Report

1. Introduction

The purpose of this practical was to set up and test a basic network using a Windows laptop and a mobile phone. The main commands used for network testing were ipconfig, ping, tracert, and arp -a. These commands helped me check IP configuration, network connectivity, the route to YouTube, and devices known to the laptop.

2. Network Setup

First, I connected my laptop to the available Wi-Fi network. I opened Command Prompt and used the following command to check the network configuration:

ipconfig

The laptop's Wi-Fi IPv4 address was: 10.125.233.182

The default gateway was: 10.125.233.172

I also checked my friend's phone and found its IPv4 address as: 192.0.0.2

The laptop and phone therefore appeared to be on different IPv4 network ranges.

3. Testing Connectivity Using Ping

I used the ping command from the laptop to test communication with the phone:

ping 192.0.0.2

Because the laptop was using the 10.125.233.x network while the phone was using 192.0.0.2, the devices were not on the same IPv4 subnet. Therefore, successful direct local communication could not be confirmed.

I also disconnected the phone from Wi-Fi and tried pinging its IP address again. The expected result was:

Request timed out.

This indicates that the phone was not reachable from the laptop.

4. Testing the Route Using Tracert

To examine the path from my laptop to YouTube, I opened Command Prompt and ran:

tracert www.youtube.com

The command displayed the different network hops between my computer and the destination. The number of hops can change depending on the ISP, location, and current Internet routing. Therefore, the hop count should be taken directly from the actual tracert output rather than using a fixed number.

5. Checking Network Devices Using ARP

I used the following command to view the ARP table:

arp -a

My actual output showed the Wi-Fi interface:

Interface: 10.125.233.182

and a dynamic gateway entry:

10.125.233.172 5e-02-07-0b-10-54 dynamic

The output also contained VMware virtual network interfaces such as 192.168.251.1 and 192.168.148.1. These are virtual adapters rather than normal home Wi-Fi devices.

6. Results and Issues Observed

The network configuration was successfully checked using ipconfig. The ping test highlighted an issue because the laptop and phone had different IPv4 network ranges. After disconnecting the phone, ping requests were expected to return "Request timed out." The tracert command was used to examine the Internet route to YouTube, while arp -a successfully displayed the laptop's known IP-to-MAC address mappings.

7. Conclusion

This practical helped me understand basic network configuration and troubleshooting. I learned how to check IP addresses using ipconfig, test connectivity using ping, trace Internet routes using tracert, and examine local ARP entries using arp -a. The practical also showed that devices should normally be connected to the same local network/subnet for direct local connectivity testing.